

DATE : 08.10.2026

DT/NT : DT

LESSON : NETWORK

SUBJECT: IP ADDRESSES

CISCO PACKET TRACER APPLICATION

BATCH : B 424-020DE

AWS-DEVOPS



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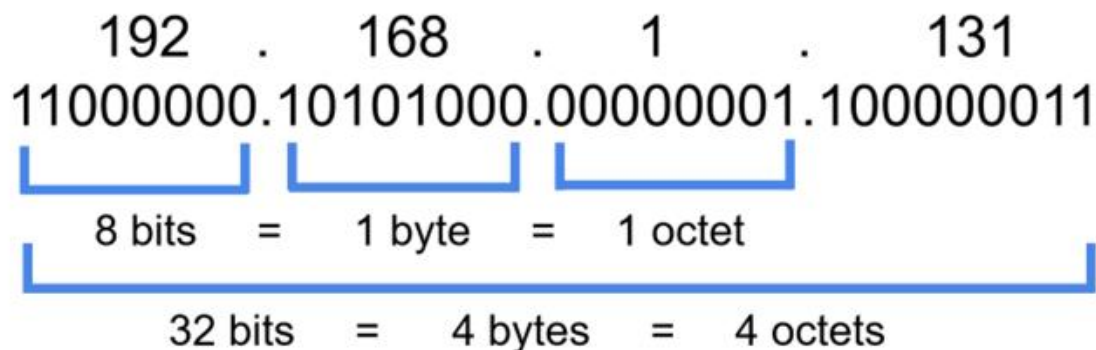


+1 (585) 304 29 59



IPv4 Address Anatomy

- Made up of 32 binary bits, which can be divided into a **network portion** and a **host portion** with the help of a subnet mask.
 - The 32 binary bits are broken into four octets (1 octet = 8 bits).
 - Each octet is converted to decimal and separated by a period (dot).
 - For this reason, an IP address is said to be expressed in dotted decimal format.



IPv4 Address Anatomy

192 . 168 . 1 . 131
11000000.10101000.00000001.100000011

8 bits = 1 byte = 1 octet

32 bits = 4 bytes = 4 octets

First Octet	Second Octet	Third Octet	Fourth Octet
192	168	1	131
11000000	10101000	00000001	100000011
8 bits	8 bits	8 bits	8 bits

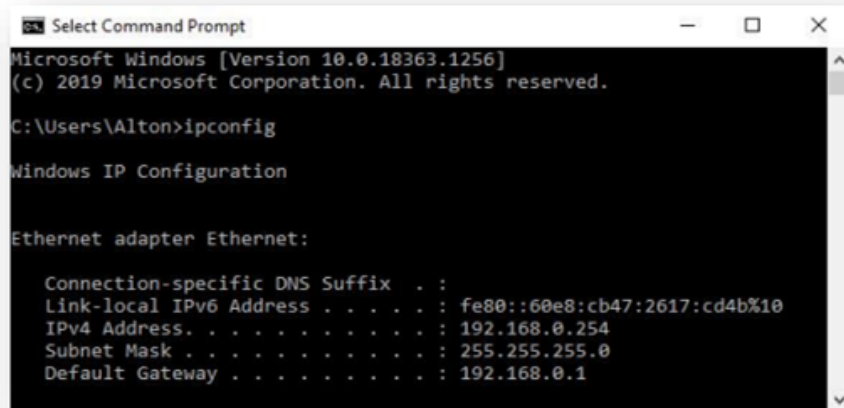
Network and Host Portion

- An IP address is broken down into two parts:
 - **Network Address**
 - Uniquely identifies each network
 - Your Street Name: 7682 **Wilshire Drive**
 - **Host Address**
 - Uniquely identifies each machine on a network
 - Your House Address: **7682** Wilshire Drive
- Network Address + Host Address = IP Address
 - **Wilshire Drive 7682**



IPv4 Address Components

- Each device on a network is assigned an IP address, subnet mask and default gateway:
 - **IP Address:** Unique logical address assigned to each device on a network.
 - **Subnet Mask:** Used by the device to determine what subnet it's on, specifically the network and host portions of the IP address.
 - **Default Gateway:** The IP address of a network's router that allows devices on the local network to communicate with other networks.



```
Select Command Prompt

Microsoft Windows [Version 10.0.18363.1256]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\Alton>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::60e8:cb47:2617:cd4b%10
    IPv4 Address. . . . . : 192.168.0.254
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1
```

Basics of Binary Math

Lecture Goals

- Convert Binary to Decimal
- Convert Decimal to Binary

Basics of Binary Math

Why is it important?

We need to know basic binary math to perform subnetting, as well as to understand how IPv4 addresses work.

Remember This

$$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$$

What is the binary 11111111 in decimal?

	128	64	32	16	8	4	2	1										
Binary	1	1	1	1	1	1	1		1									
Decimal	128	+	64	+	32	+	16	+	8	+	4	+	2	+	1	=	255	Decimal

Add the number where there is a "1".
Add zero, when there is a "0".

What is the binary 10101010 in decimal?

	128	64	32	16	8	4	2	1										
Binary	1	0	1	0	1	0	1	0										
Decimal	128	+	0	+	32	+	0	+	8	+	0	+	2	+	0	=	170	Decimal

Add the number where there is a "1".

Add zero, when there is a "0".

What is the binary 10000011 in decimal?

	128	64	32	16	8	4	2	1						
Binary	1	0	0	0	0	0	1		1					
Decimal	128	+	0	+	0	+	0	+	2	+	1	=	131	Decimal

Add the number where there is a "1".

Add zero, when there is a "0".

What's 192 in binary?

	128	64	32	16	8	4	2	1	
Binary	1	1	0	0	0	0	0		0 = 11000000
Decimal	128	+	64	+	0	+	0	+	0 = 192 Decimal

Start adding the numbers from left to right until you achieve the decimal amount you are looking for!

What's 202 in binary?

	128	64	32	16	8	4	2	1								
Binary	1	1	0	0	1	0	1		0 = 11001010							
Decimal	128	+	64	+	0	+	0	+	8	+	0	+	2	+	0 = 202	Decimal

Start adding the numbers from left to right until you achieve the decimal amount you are looking for!

What's 54 in binary?

	128	64	32	16	8	4	2	1								
Binary	0	0	1	1	0	1	1		0 = 00110110							
Decimal	0	+	0	+	32	+	16	+	0	+	4	+	2	+	0 = 54	Decimal

Start adding the numbers from left to right until you achieve the decimal amount you are looking for!

IP Address Conversion Process

192.	168.	32.	4	Dotted Decimal
11000000.	10101000.	00100000.	00000100	Binary
1 st Octet	2 nd Octet	3 rd Octet	4 th Octet	

Whether you are given an IP address in dotted-decimal or binary format, follow the respective process above for each octet one by one until you have completed the process.

CONVERSION CHART

$$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$$

1. CONVERT 11110000 TO DECIMAL

	128		64		32		16		8		4		2		1	
Binary	1		1		1		1		0		0		0		0	=
Decimal	128	+	64	+	32	+	16	+	0	+	0	+	0	+	0	= 240 Decimal

2. CONVERT 10011001 TO DECIMAL

	128		64		32		16		8		4		2		1	
Binary	1		0		0		1		1		0		0		1	=
Decimal	128	+	0	+	0	+	16	+	8	+	0	+	0	+	1	= 153 Decimal

3. CONVERT 01101011 TO DECIMAL

	128		64		32		16		8		4		2		1	
Binary	0		1		1		0		1		0		1		1	=
Decimal	0	+	64	+	32	+	0	+	8	+	0	+	2	+	1	= 107 Decimal

4. CONVERT 10110011 TO DECIMAL

	128		64		32		16		8		4		2		1	
Binary	1		0		1		1		0		0		1		1	=
Decimal	128	+	0	+	32	+	16	+	0	+	0	+	2	+	1	= 179 Decimal

5. CONVERT 240 TO BINARY

	128		64		32		16		8		4		2		1	
Binary	1		1		1		1		0		0		0		0	= 11110000 Binary
Decimal	128	+	64	+	32	+	16	+	0	+	0	+	0	+	0	=

6. CONVERT 163 TO BINARY

	128		64		32		16		8		4		2		1	
Binary	1		0		1		0		0		0		1		1	= 10100011 Binary
Decimal	128	+	0	+	32	+	0	+	0	+	0	+	2	+	1	=

7. CONVERT 94 TO BINARY

	128		64		32		16		8		4		2		1	
Binary	0		1		0		1		1		1		1		0	= 01011110 Binary
Decimal	0	+	64	+	0	+	16	+	8	+	4	+	2	+	0	=

8. CONVERT 225 TO BINARY

	128		64		32		16		8		4		2		1	
Binary	1		1		1		0		0		0		0		1	= 11100001 Binary
Decimal	128	+	64	+	32	+	0	+	0	+	0	+	0	+	1	=

9. CONVERT THE FOLLOWING IP ADDRESS FROM DECIMAL TO BINARY

192.168.98.18

- 192 = 11000000
- 168 = 10101000
- 98 = 01100010
- 18 = 00010010

Binary Format: 11000000.10101000.01100010.00010010

10. CONVERT THE FOLLOWING IP ADDRESS FROM BINARY TO DECIMAL

01000010.11010010.11000110.11000101

- 01000010 = 66
- 11010010 = 210
- 11000110 = 198
- 11000101 = 197

Dotted Decimal Format: 66.210.198.197

CONVERSION CHART

$$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$$

1. CONVERT 11110000 TO DECIMAL

	128	64	32	16	8	4	2	1	
Binary									=
Decimal	+	+	+	+	+	+	+	+	=

2. CONVERT 10011001 TO DECIMAL

	128	64	32	16	8	4	2	1	
Binary									=
Decimal	+	+	+	+	+	+	+	+	=

3. CONVERT 01101011 TO DECIMAL

	128	64	32	16	8	4	2	1	
Binary									=
Decimal	+	+	+	+	+	+	+	+	=

4. CONVERT 10110011 TO DECIMAL

	128	64	32	16	8	4	2	1	
Binary									=
Decimal	+	+	+	+	+	+	+	+	=

5. CONVERT 240 TO BINARY

	128	64	32	16	8	4	2	1	
Binary									=
Decimal	+	+	+	+	+	+	+	+	=

6. CONVERT 163 TO BINARY

	128	64	32	16	8	4	2	1	
Binary									=
Decimal	+	+	+	+	+	+	+	+	=

7. CONVERT 94 TO BINARY

	128	64	32	16	8	4	2	1	
Binary									=
Decimal	+	+	+	+	+	+	+	+	=

8. CONVERT 225 TO BINARY

	128	64	32	16	8	4	2	1	
Binary									=
Decimal	+	+	+	+	+	+	+	+	=

9. CONVERT THE FOLLOWING IP ADDRESS FROM DECIMAL TO BINARY

192.168.98.18

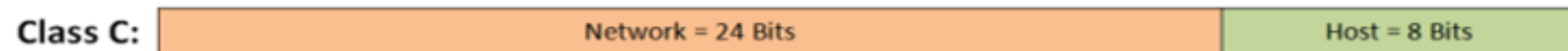
10. CONVERT THE FOLLOWING IP ADDRESS FROM BINARY TO DECIMAL

01000010.11010010.11000110.11000101

IPv4 Address Classes (Simplified)

Class	Network Bits	Host Bits	Address Range
A	8	24	1.0.0.0 – 126.255.255.255
B	16	16	128.0.0.0 – 191.255.255.255
C	24	8	192.0.0.0 – 223.255.255.255

Network and Host Bits



IPv4 Address Classes (Detailed)

Class	Leading Bits	Network Bits	Remaining Bits	Number of Networks	Hosts Per Network	Default Subnet Mask
Class A	0 (1-126)	8	24	128 (2^7)	16,777,216 (2^{24})	255.0.0.0
Class B	10 (128-191)	16	16	16,384 (2^{14})	65,536 (2^{16})	255.255.0.0
Class C	110 (192-223)	24	8	2,097,152 (2^{21})	256 (2^8)	255.255.255.0
Class D (multicast)	1110 (224-239)	Not Defined	Not Defined	Not Defined	Not Defined	Not Defined
Class E (reserved)	1111 (240-255)	Not Defined	Not Defined	Not Defined	Not Defined	Not Defined

Default Subnet Masks

- The Subnet Mask tells you which portion of the IP address identifies the network and which portion identifies the host.
- Below are default Class A, B, and C Subnet Masks.

	8 bits	8 bits	8 bits	8 bits
Class A:	Network	Host	Host	Host
IP Address	10.	0.	0.	15
Subnet Mask	11111111.	00000000.	00000000.	00000000
	255.	0.	0.	0
Class B:	Network	Network	Host	Host
IP Address	172.	16.	0	.110
Subnet Mask	11111111.	11111111.	00000000.	00000000
	255.	255.	0.	0
Class C:	Network	Network	Network	Host
IP Address	192.	168.	1.	50
Subnet Mask	11111111.	11111111.	11111111.	00000000
	255.	255.	255.	0

Let's Practice

What class are the following IP Addresses?

- **IP Address:** 9.10.40.15
- **Subnet Mask:** 255.0.0.0

- **IP Address:** 135.240.110.100
- **Subnet Mask:** 255.255.0.0

- **IP Address:** 196.200.10.5
- **Subnet Mask:** 255.255.255.0

CIDR Notation

- **CIDR:** Classless Inter-Domain Routing
 - A methodology for subnetting
 - "Slash" Notation tells you how many bits are associated with the Subnet Mask
- A shortcut way of telling us what the Subnet Mask is:
 - /8 = 11111111.00000000.00000000.00000000
 - /8 = 255.0.0.0
- 192.168.1.0 /24 = 255.255.255.0
- 10.1.0.0 /16 = 255.255.0.0
- 196.10.10.0/25 = 255.255.255.128

Understanding the Power of 2

- We use the power of 2 in IP addressing and subnetting.
- It's important to memorize the power of 2.

$2^1 = 2$	$2^2 = 4$	$2^3 = 8$	$2^4 = 16$
$2^5 = 32$	$2^6 = 64$	$2^7 = 128$	$2^8 = 256$
$2^9 = 512$	$2^{10} = 1,024$	$2^{11} = 2,048$	$2^{12} = 4,096$

Using Power of 2 to Determine Network Hosts

	8 bits	8 bits	8 bits	8 bits
Class A:	Network = 8 Bits	Hosts = 24 Bits = $2^{24} - 2 = 16,777,214$		
Class B:	Network = 16 Bits		Hosts = 16 Bits = $2^{16} - 2 = 65,534$	
Class C:	Network = 24 Bits			Hosts = 8 Bits = $2^8 - 2 = 254$

- **Hosts Per Network** = $2^h - 2$, where h is the number of host bits available.
- We subtract two because each network includes a **network address** and **broadcast address** that are not available for use by network end devices.

Public versus Private IP Addresses

Public IP Addresses

- Original Design of Internet
- “Registered” Public IP Addresses
- Assigned by an ISP to a Business or Home
- Must be Globally Unique
 - Web Servers
 - DNS Servers
 - Routers
- By the Early 1990s, the World was Running out of Public IP Addresses
- Private IP Addresses & Network Address Translation (NAT) were Born!

Private IP Addresses

- “Unregistered” – Free for Use by Anybody!
- Designed for Use within Private Internal Networks
- Can Be Used Over and Over Again
- Cannot be Used or Routed on a Public Network
- Utilizes NAT to “Speak” to Public Networks, i.e., the Internet!

Private IP Address Ranges

Class	IP Address Range	Network ID(s) (CIDR Notation)	Number of Addresses
A	10.0.0.0 – 10.255.255.255	10.0.0.0 /8 • 1 Private Class A Network	16,777,216 IP Addresses Per Network ID
B	172.16.0.0 – 172.31.255.255	172.16.0.0 – 172.31.0.0 /16 • 16 Private Class B Networks	65,534 IP Addresses Per Network ID
C	192.168.0.0 – 192.168.255.255	192.168.0.0 – 192.168.255.0 /24 • 256 Private Class C Networks	254 IP Addresses Per Network ID

The Loopback Address

- **127.0.0.0 to 127.255.255.255** is reserved for loopback, i.e., a host's own address, also known as the localhost address.
 - **127.0.0.1** is typically configured as the default loopback address on operating systems.
- Used for diagnostics purposes to check that TCP/IP is correctly installed on a host's operating system.
 - When a process creates a packet destined to the loopback address, the operating system loops it back to itself without it ever interfacing with the NIC.
 - Data sent on the loopback is forwarded by the operating system to a virtual network interface within the operating system.
- If you can successfully ping 127.0.0.1 or any IP within the loopback range, then TCP/IP on your computer is properly working.
 - Ping 127.0.0.1
 - Ping localhost
 - Ping loopback

Why Subnet?

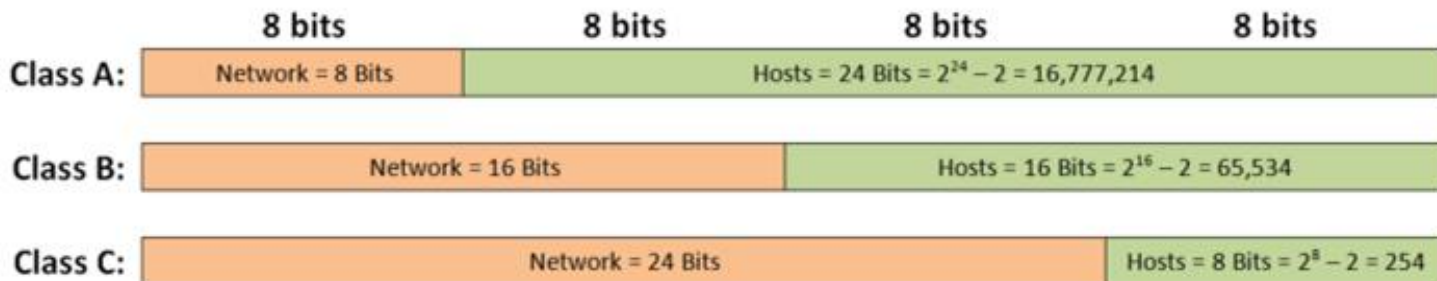
- Using default Class A, B and C subnets (called Classful IP Addressing) is inefficient:
 - Wastes unused IP Addresses (Public IP Addresses)
- Allows you to create multiple logical networks that exist within a single Class A, B, or C network.
 - Breaks up larger networks into multiple smaller sub-networks, which are called subnets
- Allows for more efficient routing via router summarization.
- Increased network security!

Fixed Length Subnetting

- We will be learning about fixed-length subnetting, known as a fixed-length subnet mask (FLSM).
- There is also variable-length subnetting (VLSM), which is beyond the scope of this beginner's course.

Process of Subnetting

- We borrow host bits to create more sub-networks (subnets) from a Class A, B, or C network.
- When you borrow hosts bits:
 - You create additional sub-networks, i.e., subnets
 - You also decrease the amount of host IP addresses available to use



How to Create Subnets

- Borrow bits from the host portion of an IP address
 - Each bit we borrow is equal to 2^1 Subnets
 - Borrow 1 Host Bit = $2^1 = 2$
 - Borrow 2 Host Bits = $2^2 = 4$
 - Borrow 3 Host Bits = $2^3 = 8$
 - Borrow 4 Host Bits = $2^4 = 16$
 - Etc.

Creating Subnets Visualized

Default Class C Network (8 Host Bits)



2 Host Bits Borrowed = 2^2 = Subnetted into 4 Subnets

1	2	3	4
---	---	---	---



3 Host Bits Borrowed = 2^3 = Subnetted into 8 Subnets

1	2	3	4	5	6	7	8
---	---	---	---	---	---	---	---

Subnet Calculation Table (2^x)

Host Bits Borrowed	2 ^x	Number of Subnets Created
1	2 ¹	2
2	2 ²	4
3	2 ³	8
4	2 ⁴	16
5	2 ⁵	32
6	2 ⁶	64
7	2 ⁷	128
8	2 ⁸	256
9	2 ⁹	512
10	2 ¹⁰	1,024
11	2 ¹¹	2,048
12	2 ¹²	4,096
Etc....		

Subnet Hosts & Addresses Calculation Table (2^y)

Host Bits Left	2^y	Addresses per Subnet (2^y)	Hosts per Subnet ($2^y - 2$)
1	2^1	2	0
2	2^2	4	2
3	2^3	8	6
4	2^4	16	14
5	2^5	32	30
6	2^6	64	62
7	2^7	128	126
8	2^8	256	254
9	2^9	512	510
10	2^{10}	1,024	1,022
11	2^{11}	2,048	2,046
12	2^{12}	4,096	4,094

Subnetting Reference Tables

POWER OF 2'S TABLE

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32$$

$$2^6 = 64$$

$$2^7 = 128$$

$$2^8 = 256$$

$$2^9 = 512$$

$$2^{10} = 1,024$$

$$2^{11} = 2,048$$

$$2^{12} = 4,096$$

$$2^{13} = 8,192$$

$$2^{14} = 16,384$$

$$2^{15} = 32,768$$

$$2^{16} = 65,536$$

DEFAULT SUBNET MASK

Class	Format	Default Subnet Mask
A	network. host . host . host	255.0.0.0
B	network.network. host . host	255.255.0.0
C	network.network.network. host	255.255.255.0

BINARY MATH TABLE

128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1



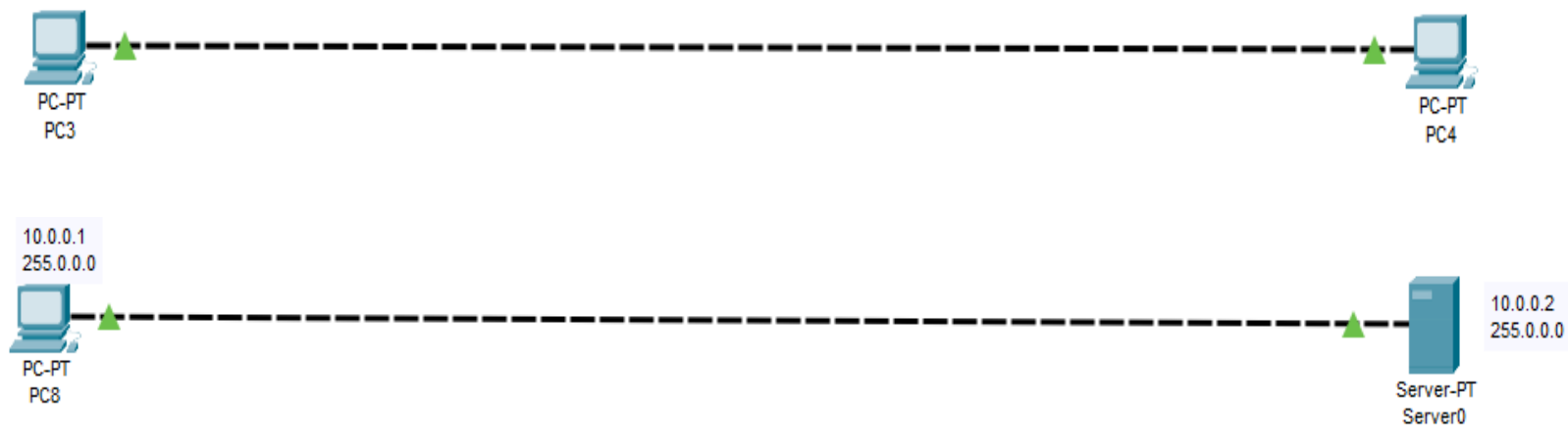
Example:

**Do the subnetting operation for 5 departments using the IP address below:
195.168.10.0/24**

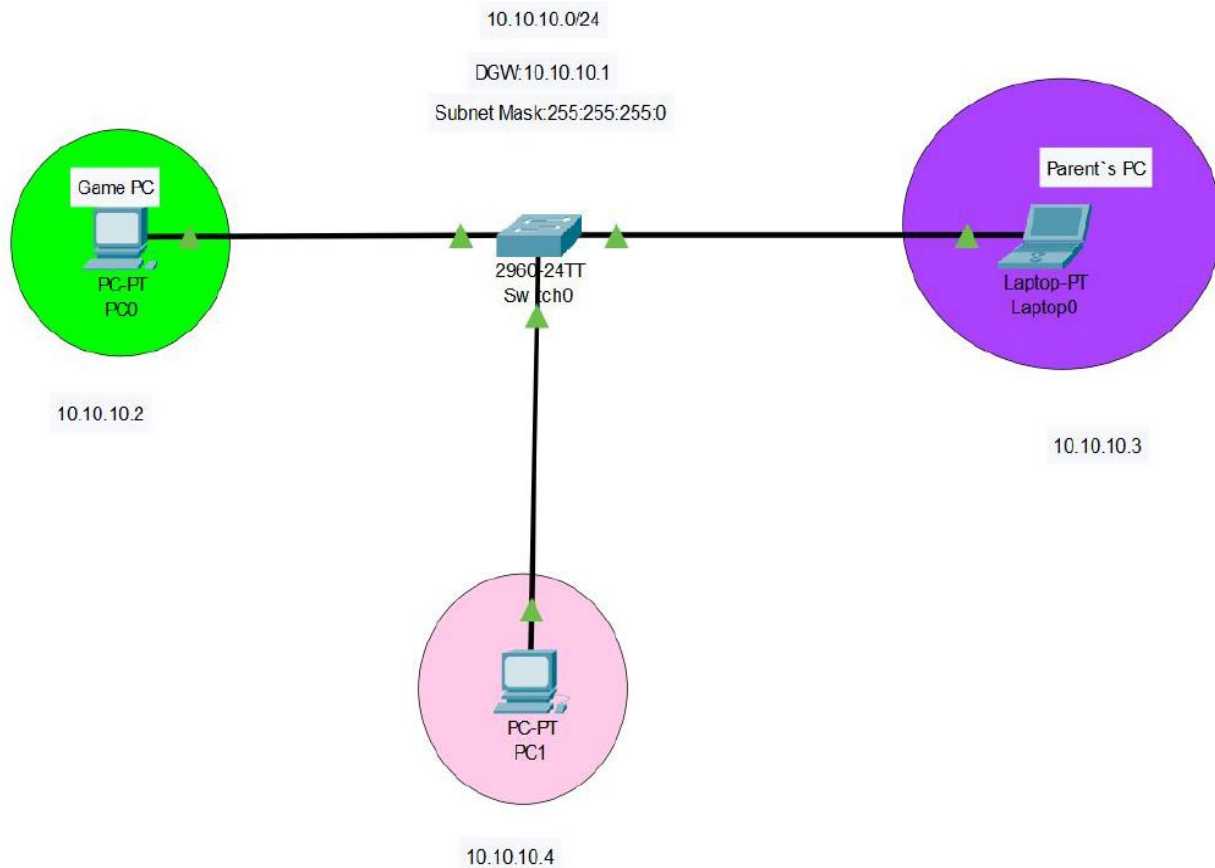
Subnet #	Network Address	First Usable IP	Last Usable IP	Broadcast Address
1	195.168.10.0/27	195.168.10.1	195.168.10.30	195.168.10.31
2	195.168.10.32/27	195.168.10.33	195.168.10.62	195.168.10.63
3	195.168.10.64/27	195.168.10.65	195.168.10.94	195.168.10.95
4	195.168.10.96/27	195.168.10.97	195.168.10.126	195.168.10.127
5	195.168.10.128/27	195.168.10.129	195.168.10.158	195.168.10.159

Cisco Packet Tracer Application

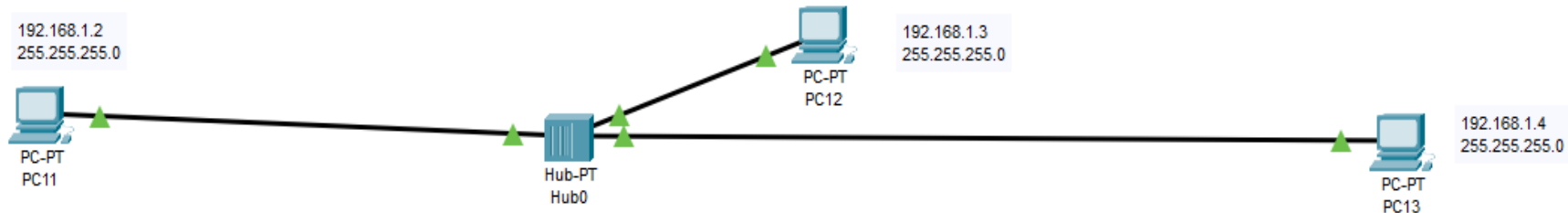
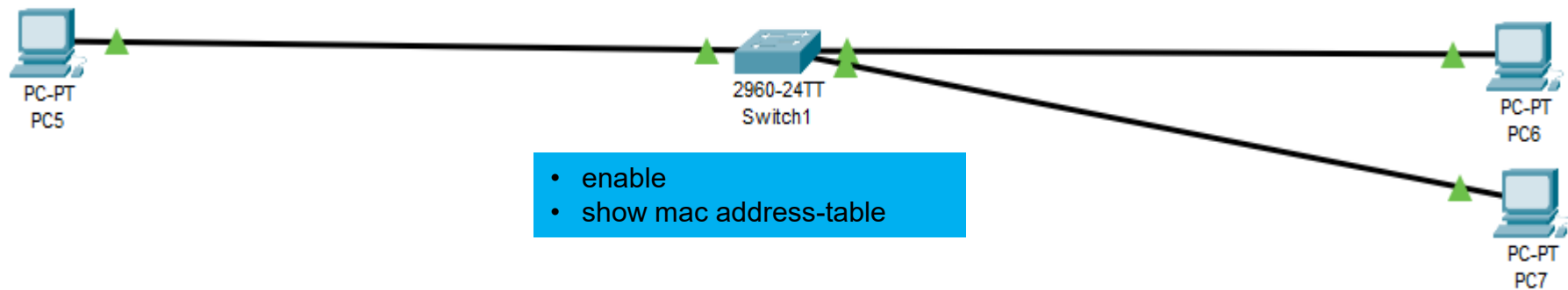
Example 1

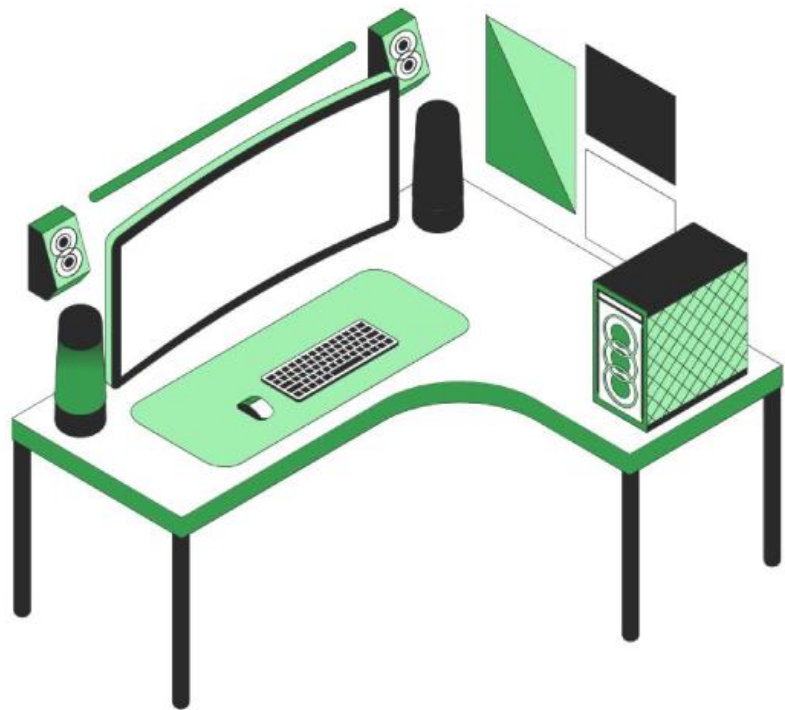


Example 2



Example 3





Do you have any questions?

Send it to us! We hope you learned something new.